Notation

This page is a reference, not a section: there is nothing to work through here, and it carries no exercises and no quick checks. Use it when you meet a symbol and want to know what it means and where the book fixed it. The symbols are grouped by the kind of thing they name, not by chapter, because a reader arrives with a symbol and not with a chapter number. Every entry links to the place where the book defines it; follow the link for the hypotheses, which a one-line gloss cannot carry.

Read the conventions first. Most of what looks like a clash between two symbols is resolved by one of them.

24.-1.1 The standing conventions

The field. F is an arbitrary field throughout, and $\mathbb{F}_p$ is the field with p elements (Definition 0.7.1). Chapters 11 to 13 and much of Part V need $\mathbb{R}$ or $\mathbb{C}$, and say so where they do. Vector spaces are V, U, W, plain italic, and are finite-dimensional unless the statement says otherwise.

Bold is for matrices and coordinate vectors, never for maps, spaces, or abstract vectors. A matrix is a bold upright capital, $\mathbf{A}$; a coordinate vector, an element of F^n, is a bold lowercase letter, $\mathbf{v}$; an element of an abstract vector space is plain italic, v, so the letter shows whether a computation is in coordinates or in the structure. The test is exact: an object is bold only when it is an element of a fixed, finite-dimensional F^n or $F^{m\times n}$ viewed in coordinates. So the zero matrix is bold, $\mathbf{O}$, as every matrix is; the zero vector is bold in F^n and plain, 0, in an abstract space; and a sequence in $F^{\mathbb{N}}$, a coset $v + U$, and the quaternion units i, j, k are plain. A coset's representative keeps its own type, so in F^n/U one writes $\mathbf{x} + U$; an operator or linear map is plain italic, T; a space is plain italic, V. Entries stay plain: a_{ij}, or $(\mathbf{A})_{ij}$. A Greek letter naming a matrix is bold too, as $\mathbf{\Lambda}$ or $\mathbf{\Sigma}$.

Maps and composition. Linear maps are $T, S\colon V \to W$, never f. Composition is written ST, and it means: first T, then S. For functions in general the book writes $g \circ f$.

Coordinates. $[v]_{\mathcal{B}}$ is the coordinate vector of v in the ordered basis $\mathcal{B}$ (Definition 1.5.2), and $[T]_{\mathcal{B}}^{\mathcal{C}}$ is the matrix of T with input basis $\mathcal{B}$ and output basis $\mathcal{C}$ (Definition 4.1.1). The subscript is the basis read into the first slot and the superscript the basis read into the second: for a map that is the input basis below and the output basis above, and for a bilinear form $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ it is the basis used on the first argument below and the one used on the second above. Both slots are always written out, even when they hold the same basis. A basis is an **ordered** list, written in parentheses.

Transpose and adjoint. $\mathbf{A}^\top$ is the transpose (Definition 0.9.10) and $\mathbf{A}^*$ the conjugate transpose (Definition 11.1.2), which is also the symbol for the adjoint of an operator (Definition 11.6.1). The prime T' is the dual map (Definition 5.4.1), and never a derivative of a map.

Inner products are linear in the first slot and conjugate-linear in the second (Definition 11.1.1). Over $\mathbb{R}$ the distinction disappears.

Ordering of eigenvalues and singular values. For Hermitian $\mathbf{A}$, the eigenvalues are listed **decreasing**, $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$, with multiplicity. Singular values are listed decreasing as well, $\sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq 0$ (Definition 13.9.1). Where a proof wants the increasing order it says so.

Indices start at 1, with three announced exceptions: Chapter 12 §09, where circulant and Fourier subscripts run from 0 to $n-1$ and are read modulo n; Chapter 22 §§06 to 10, where the coordinates of F^{n+1} run from 0 so that $\mathbb{P}^n(F)$ has dimension n (Definition 22.6.2); and Chapter 23 §09, which follows Chapter 12 §09 for group elements and characters.

Defining, not asserting. $\coloneqq$ means "is defined to be". A plain $=$ asserts an equality that has been or will be proved.

24.-1.2 Numbers, sets, and fields

Symbol	Meaning	Defined at
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	natural numbers (with 0), integers, rationals, reals, complex numbers	Definition 0.3.1
F, $\mathbb{F}_p$	a field; the field with p elements	Definition 0.7.1
$\operatorname{char} F$	characteristic of a field: the least $n > 0$ with $n \cdot 1 = 0$, or 0	Definition 0.7.2
$\overline{z}$, $\lvert z \rvert$	complex conjugate and modulus	Definition 0.6.2
$\{x \in S : P(x)\}$	set-builder notation, with a colon	Definition 0.3.2
$\emptyset$, $S \subseteq T$	the empty set; inclusion	Definition 0.3.3, Definition 0.3.4
$S \cup T$, $S \cap T$, $S \setminus T$, S^c	union, intersection, difference, complement	Definition 0.3.6, Definition 0.3.7, Definition 0.3.8
$S \times T$, $(x_1, \ldots, x_n)$	Cartesian product and n-tuples	Definition 0.3.10, Definition 0.3.11
$\mathcal{P}(S)$	power set, the set of all subsets	Definition 0.3.12
$f\colon X \to Y$, $x \mapsto f(x)$	a function and its rule	Definition 0.4.1
$f(A)$, $f^{-1}(B)$	image of a subset, preimage of a subset	Definition 0.4.2
$g \circ f$, id_X, f^{-1}	composition, identity function, inverse function	Definition 0.4.4, Definition 0.4.5, Definition 0.4.7
$x \sim y$, $[x]$, $X/\!\sim$	equivalence relation, equivalence class, quotient set	Definition 0.5.2, Definition 0.5.3, Definition 0.5.5
S_n, $\operatorname{sgn}(\sigma)$, $\operatorname{inv}(\sigma)$	symmetric group; sign of a permutation; number of inversions	Example (Groups), Definition 7.3.2, Definition 7.3.1
A_n	alternating group, the even permutations	Definition 7.3.3
$F[x]$, $F[x]_{\le n}$	polynomials over F; those of degree at most n	Definition 0.8.2, Definition 0.8.4
$\deg p$, $p(c)$	degree; evaluation at a scalar	Definition 0.8.3, Definition 0.8.6
$p \mid q$, $\gcd(p, q)$, $\operatorname{lcm}(p, q)$	divisibility, greatest common divisor, least common multiple in $F[x]$	Definition 6.1.1, Definition 6.2.2, Definition 6.2.4
$\langle p \rangle$	the ideal of $F[x]$ generated by p: all multiples of p	Definition 6.2.1
$\operatorname{mult}_c(f)$	multiplicity of c as a root of f	Definition 6.4.1

Symbol	Meaning	Defined at
p'	formal derivative of a polynomial	Definition 6.4.3
ℓ_i	Lagrange basis polynomial, $\ell_i(c_j) = \delta_{ij}$	Definition 6.6.1

24.-1.3 Spaces

Symbol	Meaning	Defined at
V, U, W	vector spaces over F; U is usually a subspace	Definition 1.1.1, Definition 1.2.1
F^n, F^S	column vectors; **all** functions $S \to F$. An element of F^S, a sequence in $F^{\mathbb{N}}$ included, is plain	Definition 1.1.1
$F^{(S)}$	the free vector space on S: the **finitely supported** functions $S \to F$	Definition 15.2.3
$M_{m\times n}(F)$, $M_n(F)$	matrices over F	Definition 0.9.1
$\operatorname{span}(S)$	span; the span of the empty set is $\{0\}$	Definition 1.3.2
$\dim V$, $\dim_F V$	dimension, over F when the field matters	Definition 1.6.1
$\mathcal{B} = (v_1, \ldots, v_n)$	an **ordered** basis; $\mathcal{E}$ is the standard basis	Definition 1.5.1, Example (Standard Bases)
$[v]_{\mathcal{B}}$	coordinate vector of v in $\mathcal{B}$	Definition 1.5.2
$U + W$, $U \oplus W$	sum of subspaces; internal direct sum	Definition 1.7.1, Definition 1.7.2
$V \times W$	product of vector spaces (the external direct sum)	Definition 4.4.1
$v + U$, V/U	coset, plain even in F^n/U (its representative keeps its own type: $\mathbf{x} + U$); quotient space	Definition 4.4.2, Definition 4.4.3
$\operatorname{codim} U$	codimension, $\dim V/U$	Definition 5.5.1
V^*, φ	dual space; a linear functional, always a Greek letter	Definition 5.1.2, Definition 5.1.1
$\varphi_1, \ldots, \varphi_n$	the dual basis of $\mathcal{B}$	Definition 5.1.3
U^0	annihilator of U inside V^*	Definition 5.2.1
V^{**}, ev_v	double dual; evaluation at v	Definition 5.3.1
$V_{\mathbb{C}}$	complexification of a real space	Definition 1.9.2
$\mathcal{F}$	a family of operators in $\mathcal{L}(V)$	Definition 9.9.1

24.-1.4 Vectors, matrices, and their operations

Symbol	Meaning	Defined at
$\mathbf{v}, \mathbf{u}, \mathbf{w}, \mathbf{x}, \mathbf{y}$	coordinate vectors, in F^n: bold lowercase	Definition 1.1.2
v, u, w, x, y	vectors of an abstract space V: plain italic	Definition 1.1.1
$\mathbf{0}$, 0, $\mathbf{1}$	the zero vector of F^n, bold, and of an abstract space, plain; the all-ones vector	Definition 1.1.2, Definition 9.11.1

Symbol	Meaning	Defined at
$\mathbf{e}_1, \ldots, \mathbf{e}_n$	the standard basis of F^n	Example (Standard Bases)
$\mathbf{E}_{ij}$	matrix unit: 1 in entry (i, j), zeros elsewhere	Example (Standard Bases)
$\mathbf{A}, \mathbf{B}, \mathbf{C}$, a_{ij}	matrices, bold upright capitals; their entries, plain	Definition 0.9.1
$\mathbf{O}_{m\times n}$, $\mathbf{O}_n$, $\mathbf{I}$, $\mathbf{I}_n$	zero matrix, a bold O like every matrix ($\mathbf{O}_n$ when square), also as a block; identity matrix	Definition 0.9.2, Definition 0.9.3
δ_{ij}	Kronecker delta: 1 when $i = j$, else 0	Definition 0.9.3
$\operatorname{diag}(d_1, \ldots, d_n)$	the diagonal matrix with these diagonal entries	Definition 0.9.4
$\mathbf{A}^\top$, $\mathbf{A}^*$	transpose; conjugate transpose	Definition 0.9.10, Definition 11.1.2
$\operatorname{tr}\mathbf{A}$	trace, the sum of the diagonal entries	Definition 0.9.12
$\det\mathbf{A}$	determinant	Definition 7.4.1
$\operatorname{adj}\mathbf{A}$	adjugate, the transposed matrix of cofactors	Definition 7.6.2
M_{ij}, C_{ij}	minor and cofactor $(-1)^{i+j}M_{ij}$	Definition 7.6.1
$\mathbf{A}^{-1}$	inverse of a nonsingular matrix	Definition 0.9.13
$p(\mathbf{A})$	a polynomial evaluated at a matrix	Definition 0.9.14
$\mathbf{P}_{ij}$, $\mathbf{D}_i(c)$, $\mathbf{I} + c\mathbf{E}_{ij}$	the three elementary matrices: swap, scale, add	Definition 3.4.1
$\mathbf{P}_\sigma$	permutation matrix	Definition 3.4.2
$\mathbf{A} = \text{Ł}\mathbf{U}$, $\mathbf{PA} = \text{Ł}\mathbf{U}$	the LU and PLU factorizations	Definition 3.6.1, Definition 3.6.2
$\operatorname{col}(\mathbf{A})$, $\operatorname{row}(\mathbf{A})$, $\operatorname{nul}(\mathbf{A})$	column space, row space, null space	Definition 3.5.1, Definition 3.5.2, Definition 3.5.3
$\operatorname{rank}\mathbf{A}$, $\operatorname{nullity}\mathbf{A}$	rank and nullity of a matrix	Definition 3.5.4, Definition 3.5.5
$\mathbf{A} \sim \mathbf{B}$	similar: $\mathbf{B} = \mathbf{P}^{-1}\mathbf{AP}$	Definition 4.2.2
$\mathbf{A} \simeq \mathbf{B}$	congruent: $\mathbf{B} = \mathbf{P}^\top\mathbf{AP}$	Definition 14.2.1
$\mathbf{A} \approx \mathbf{B}$	equivalent over $F[x]$: $\mathbf{B} = \mathbf{UAV}$ with $\mathbf{U}, \mathbf{V}$ invertible over $F[x]$	Definition 10.7.2
equivalent matrices	$\mathbf{B} = \mathbf{QAP}$ with $\mathbf{P}, \mathbf{Q}$ nonsingular	Definition 4.3.1
$\mathbf{A} \oplus \mathbf{B}$	block diagonal matrix with the given square diagonal blocks	Definition 8.1.2
$\mathbf{M}/\mathbf{A}$, $\mathbf{M}/\mathbf{D}$	Schur complement of an invertible block of a block matrix	Definition 8.3.1
$\mathbf{A} \otimes \mathbf{B}$	Kronecker product	Definition 8.4.1
$\operatorname{vec}\mathbf{X}$	vectorization: the columns of $\mathbf{X}$ stacked into one column	Definition 8.4.2
$\mathbf{A} \circ \mathbf{B}$	Hadamard (entrywise) product	Definition 13.8.1
$[\mathbf{A}, \mathbf{B}] = \mathbf{AB} - \mathbf{BA}$	commutator	Definition 8.5.1
$\mathbf{J}$	the all-ones matrix, every entry 1	Example (Orthogonally diagonalizing a 3 by 3 matrix)

Symbol	Meaning	Defined at
$\mathbf{A}'(t)$, $\mathbf{x}'(t)$	entrywise derivative of a matrix- or vector-valued function of a real variable	Definition 16.6.1
$e^{\mathbf{A}}$, $f(\mathbf{A})$	matrix exponential; a function of a matrix through its Jordan form	Definition 10.9.2, Definition 10.9.1
$\lim_m \mathbf{A}^m$	entrywise convergence of matrices	Definition 10.10.1
$\mathbf{Ax} = \mathbf{b}$, $\mathbf{Ax} = \mathbf{0}$	a linear system and its homogeneous version	Definition 3.1.1, Definition 3.1.2
echelon form	the staircase shapes elimination produces	Definition 3.2.3, Definition 3.2.4

24.-1.5 Maps and their spaces

Symbol	Meaning	Defined at
$T, S\colon V \to W$	linear maps; ST means first T, then S	Definition 2.1.1
$T_{\mathbf{A}}$	the map $F^n \to F^m$, $\mathbf{x} \mapsto \mathbf{Ax}$	Example (Multiplication by a Matrix)
$\ker T$, $\operatorname{im} T$	kernel and image	Definition 2.2.1, Definition 2.2.2
$\operatorname{rank} T$, $\operatorname{nullity} T$	$\dim \operatorname{im} T$ and $\dim \ker T$	Definition 2.3.2, Definition 2.3.1
$\mathcal{L}(V,W)$, $\mathcal{L}(V)$	all linear maps $V \to W$; the operators on V	Definition 2.4.1
$\operatorname{End}(V)$	the same space $\mathcal{L}(V)$, named as an **algebra** under composition	Definition 23.1.1
$p(T)$, id_V	a polynomial in an operator; the identity operator	Definition 2.4.2, Definition 2.1.1
$[T]_{\mathcal{B}}^{\mathcal{C}}$	matrix of T from the input basis $\mathcal{B}$ to the output basis $\mathcal{C}$	Definition 4.1.1
$[\mathrm{id}]_{\mathcal{B}}^{\mathcal{C}}$	change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$	Definition 4.2.1
$V \cong W$	isomorphic vector spaces	Definition 2.5.2, Definition 2.5.3
T'	dual map $W^* \to V^*$; the star is kept for adjoints	Definition 5.4.1
$\operatorname{tr} T$, $\det T$	trace and determinant of an operator	Definition 4.5.2, Definition 7.5.1
$T\|_U$, $\bar{T}$	restriction of T to an invariant subspace; the operator it induces on V/U	Definition 9.1.1, Definition 9.1.2
$\mathrm{GL}_n(F)$, $\mathrm{SL}_n(F)$	general and special linear groups	Example (Groups), Theorem 7.5.9
$\mathcal{M}(V_1, \ldots, V_k; W)$	multilinear maps $V_1 \times \cdots \times V_k \to W$; the semicolon separates inputs from output	Definition 15.1.2

24.-1.6 Inner products and norms

Symbol	Meaning	Defined at
$\langle u, v\rangle$	inner product: **linear in the first slot**, conjugate-linear in the second	Definition 11.1.1
$\langle \mathbf{A}, \mathbf{B}\rangle = \operatorname{tr}(\mathbf{B}^*\mathbf{A})$	the Frobenius inner product on matrices	Definition 11.1.1
$\|v\|$	norm; the induced norm $\sqrt{\langle v, v\rangle}$ unless said otherwise	Definition 11.1.3, Definition 16.1.1
$u \perp v$, θ	orthogonal vectors; the angle between two nonzero real vectors	Definition 11.1.4, Definition 11.1.5
$\mathbf{G}$	Gram matrix of a list, $(\mathbf{G})_{ij} = \langle v_j, v_i\rangle$	Definition 11.2.3
$U^\perp$	orthogonal complement	Definition 11.3.1
P_U, $d(v, U)$	orthogonal projection onto U; distance to U	Definition 11.3.2, Definition 11.3.3
T^*	adjoint of an operator	Definition 11.6.1
self-adjoint, normal	$T^* = T$; $T^*T = TT^*$	Definition 11.6.2, Definition 11.6.3
$\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$	orthogonal, special orthogonal, and unitary groups	Definition 11.7.3
$\mathbf{H}_\mathbf{w}$, $\mathbf{G}(i, j; c, s)$	Householder reflection; Givens rotation	Definition 11.8.1, Definition 11.8.2
P_n, T_n	Legendre and Chebyshev polynomials	Definition 11.10.1, Definition 24.7.4
$\|\mathbf{x}\|_p$	the p-norm; q is the conjugate exponent, $1/p + 1/q = 1$	Definition 16.1.1, Corollary 18.11.5
$\|f\|_\infty$	sup norm on $C[0, 1]$	Definition 16.1.1
$B_{\|\cdot\|}$, B_1, B_2, B_∞	closed unit ball of a norm; the balls of the three p-norms	Definition 16.1.2
$\|\mathbf{A}\|_F$	Frobenius norm, $\sqrt{\operatorname{tr}(\mathbf{A}^*\mathbf{A})}$; the subscript is obligatory	Lemma 11.7.5
$\|\mathbf{A}\|$, $\|\mathbf{A}\|_1$, $\|\mathbf{A}\|_\infty$	operator norm induced by a vector norm; largest column sum; largest row sum	Definition 16.3.2
$\kappa(\mathbf{A}) = \|\mathbf{A}\|\|\mathbf{A}^{-1}\|$	condition number	Definition 16.8.1
$\|\mathbf{y}\|_*$	dual norm, $\max_{\|\mathbf{x}\|\le 1}\langle \mathbf{x}, \mathbf{y}\rangle$	Definition 18.4.4
$\|\!\|\!\|\mathbf{A}\|\!\|\!\|$, $\|\!\|\!\|\mathbf{A}\|\!\|\!\|_{(k)}$	unitarily invariant norm; the Ky Fan k-norm $\sigma_1 + \cdots + \sigma_k$	Definition 21.4.1, Definition 21.4.2
Φ, Φ^*	symmetric gauge function and its dual, with $\|\!\|\!\|\mathbf{A}\|\!\|\!\| = \Phi(\sigma(\mathbf{A}))$	Definition 21.3.1, Definition 21.3.3
$\langle \mathbf{x}, \mathbf{y}\rangle_\mathbf{A}$, $\|\mathbf{x}\|_\mathbf{A}$	the inner product $\mathbf{y}^*\mathbf{A}\mathbf{x}$ of a positive definite $\mathbf{A}$, and its energy norm	Theorem 13.4.2, Definition 24.7.1
$\Theta(U, W)$, $\theta_1 \le \cdots \le \theta_k$	principal angles between two subspaces, in $[0, \pi/2]$	Definition 20.8.1

24.-1.7 Bilinear forms, tensors, and exterior algebra

Symbol	Meaning	Defined at
$\beta\colon V \times V \to F$, $q(v) = \beta(v, v)$	bilinear form; the quadratic form it carries	Definition 14.1.1, Definition 14.3.2
$[\beta]_{\mathcal{B}}^{\mathcal{B}}$	Gram matrix of a form, $\beta(v_i, v_j)$, indices uncrossed; $\mathcal{B}$ is written twice because each slot of the form is given a basis	Definition 14.1.2
$\operatorname{rad}(\beta)$	radical of a form; β is nondegenerate exactly when it is $\{0\}$	Definition 14.1.3, Definition 14.1.4
discriminant	the square class of $\det[\beta]_{\mathcal{B}}^{\mathcal{B}}$ in $F^\times$, the same for every basis	Definition 14.2.2
(n_+, n_-, n_0)	inertia of a real symmetric form; the signature is the pair (n_+, n_-)	Definition 14.5.1
$\operatorname{In}(\mathbf{A})$	inertia triple of a Hermitian matrix, by sign of eigenvalue	Definition 17.10.1
$\mathbf{H}_f(\mathbf{a})$	Hessian matrix of a twice-differentiable function at a point	Definition 14.6.2
isotropic, totally isotropic	$q(v) = 0$ with $v \neq 0$; a subspace on which β vanishes	Definition 14.7.1, Definition 14.7.2
hyperbolic plane, $\operatorname{ind}(\beta)$	the plane of a hyperbolic pair; the Witt index	Definition 14.7.3, Definition 14.7.4
$U \perp W$	orthogonal direct sum for a form: the summands pair to zero	Definition 14.7.3
$(V, \beta) \cong (V', \beta')$	**isometric** bilinear spaces (this use of $\cong$ is local to forms)	Definition 14.8.1
$\operatorname{Isom}(\beta)$	isometry group of a form, $\{\mathbf{P} : \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}\}$	Definition 14.10.1
$\mathrm{O}(p, q)$, $\mathbf{I}_{p,q}$, $\mathrm{Sp}(2m, F)$, $\mathbf{\Omega}_{2m}$	the indefinite orthogonal and symplectic groups and their standard matrices	Definition 14.10.2
$V \otimes W$, $v \otimes w$	tensor product and a simple tensor	Definition 15.2.1, Definition 15.2.2
$S \otimes T$, $\mathcal{B} \otimes \mathcal{C}$	tensor product of maps; the product basis, first factor slow	Definition 15.3.1, Definition 15.3.2
Θ	the isomorphism $V^* \otimes W \to \mathcal{L}(V, W)$	Theorem 15.3.9
C, C_b^a	contraction $V^* \otimes V \to F$; contraction of one upper against one lower slot	Definition 15.4.1, Definition 15.4.3
$T_q^p(V)$, $T^{i_1 \ldots i_p}_{j_1 \ldots j_q}$	mixed tensors of type (p, q) and their components	Definition 15.4.2, Definition 15.5.1
$V^{\otimes k}$, $\mathrm{T}(V)$	k-th tensor power, with $V^{\otimes 0} = F$; the tensor algebra	Definition 15.6.1, Definition 15.6.2
$\operatorname{Sym}^k V$, $\operatorname{Sym}(V)$	k-th symmetric power; the symmetric algebra, written by juxtaposition	Definition 15.7.2, Definition 15.7.3
$(V^{\otimes k})^{S_k}$	the symmetric tensors, fixed by every permutation of the slots	Definition 15.7.5
$F[x_1, \ldots, x_n]$	the polynomial algebra in n variables	Definition 15.7.4
$\Lambda^k V$, $v_1 \wedge \cdots \wedge v_k$	k-th exterior power and a wedge	Definition 15.8.2
ΛV, e_S	exterior algebra; its basis element for an increasing index set	Definition 15.10.1
$\Lambda^k T$	exterior power of a linear map; on the top power it is $\det T$	Definition 15.9.1

Symbol	Meaning	Defined at
ε_v	wedging by a fixed vector, $\omega \mapsto v \wedge \omega$	Lemma 15.9.1
$\mathcal{I}$, $\mathcal{I}_q$	a two-sided ideal of an algebra; the one whose quotient is the Clifford algebra	Definition 15.10.3, Definition 14.12.2
$\mathbb{H}$, i, j, k	the quaternions; its units, plain like the complex i	Definition 14.12.3

24.-1.8 Spectra and canonical forms

Symbol	Meaning	Defined at
$p_T(x) = \det(x\mathbf{I} - T)$	characteristic polynomial, monic	Definition 7.10.1, Definition 7.10.2
$m_T(x)$	minimal polynomial: the monic generator of the annihilating ideal	Definition 9.5.1, Definition 6.7.1
$\operatorname{spec}(T)$	spectrum, the set of eigenvalues	Definition 9.2.3
$E_\lambda(T) = \ker(T - \lambda I)$	eigenspace	Definition 9.2.1, Definition 9.2.2
$a_T(\lambda)$, $g_T(\lambda)$	algebraic and geometric multiplicity of λ	Definition 9.3.1, Definition 9.3.2
$G_\lambda(T)$	generalized eigenspace, $\ker(T - \lambda I)^n$	Definition 10.1.1, Definition 10.1.2
$\mathbf{J}_k(\lambda)$	$k \times k$ Jordan block: λ on the diagonal, 1 on the **super**diagonal	Definition 10.3.1, Definition 10.3.2
$\mathbf{\Lambda}(\lambda)$, $\mathbf{C}_k(\lambda)$	rotation-scaling block of a conjugate pair; the real Jordan block built from it	Definition 10.4.1
$\mathbf{C}(p)$	companion matrix of a monic p	Definition 10.5.3
$Z(v;T)$, $m_{T,v}$	cyclic subspace generated by v; the T-annihilator of v	Definition 10.5.1, Definition 10.5.2
$d_1 \mid d_2 \mid \cdots \mid d_r$	invariant factors, with $d_r = m_T$; their prime powers are the elementary divisors	Definition 10.6.1
$\rho(\mathbf{A})$	spectral radius, the largest modulus in the spectrum	Definition 16.4.1
$\mathbf{\Pi}$	the limit $\lim_m \mathbf{A}^m$ of a convergent matrix	Theorem 10.10.4
$\mathbf{A} \succeq \mathbf{O}$, $\mathbf{A} \succ \mathbf{O}$	positive semidefinite, positive definite; Hermitian is implied	Definition 13.1.1
$\mathbf{A} \succeq \mathbf{B}$	the Loewner order, $\mathbf{A} - \mathbf{B} \succeq \mathbf{O}$	Definition 13.5.1
$\mathbf{A}^{1/2}$	the unique positive semidefinite square root	Theorem 13.2.1
$\lvert\mathbf{A}\rvert = (\mathbf{A}^*\mathbf{A})^{1/2}$	absolute value of a matrix	Definition 13.2.1
$\sigma_1(\mathbf{A}) \geq \cdots \geq \sigma_{\min(m,n)}(\mathbf{A})$	singular values, decreasing	Definition 13.9.1
$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$	a singular value decomposition: $\mathbf{U}$ and $\mathbf{V}$ unitary, $\mathbf{\Sigma}$ the same size as $\mathbf{A}$ with the singular values on its diagonal and zeros elsewhere	Theorem 13.9.1
$\mathbf{A}^+$	Moore–Penrose pseudoinverse	Definition 13.12.1
$\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$, $\boldsymbol{\lambda}(\mathbf{A})$	eigenvalues of a Hermitian matrix, decreasing; the vector of them	Definition 17.1.1, Definition 17.8.1

Symbol	Meaning	Defined at
$R_{\mathbf{A}}(\mathbf{x})$, $R_T(v)$	Rayleigh quotient $\langle \mathbf{Ax}, \mathbf{x}\rangle / \langle \mathbf{x}, \mathbf{x}\rangle$	Definition 17.1.1
$s_k(\mathbf{A})$	Ky Fan partial sum $\lambda_1(\mathbf{A}) + \cdots + \lambda_k(\mathbf{A})$	Corollary 17.6.5
$\mathbf{d}(\mathbf{A})$	the diagonal vector $(a_{11}, \ldots, a_{nn})$ of a Hermitian matrix	Definition 17.8.1
$\mathcal{W} = (W_1, \ldots, W_k)$	a flag of subspaces, and a list adapted to it	Definition 17.7.1
$\mathcal{H}(\mathbf{A})$	Hermitian dilation, whose eigenvalues are $\pm\sigma_i(\mathbf{A})$ and zeros	Proposition 17.9.3
$r_i(\mathbf{A})$, $D_i(\mathbf{A})$	Gershgorin radius and disc	Definition 20.1.1
$O_{ij}(\mathbf{A})$	Brauer oval of Cassini	Theorem 20.2.1
$W(\mathbf{A})$	numerical range, $\{\mathbf{x}^*\mathbf{Ax} : \|\mathbf{x}\|_2 = 1\}$	Definition 20.2.1
$\mathbf{E}$, $\widetilde{\mathbf{A}}$	a perturbation, in $\mathbf{A} + \mathbf{E}$; a perturbed matrix when it is not of that shape	Definition 16.8.1
$\mathrm{sep}_F(\mathbf{A}, \mathbf{B})$	separation of two matrices in the Frobenius norm	Definition 20.7.1
$\kappa_{\mathbf{A}}(\lambda)$	condition number of a simple eigenvalue	Definition 20.11.1
$\mathcal{S}_{\mathbf{A},\mathbf{B}}$	Sylvester operator $\mathbf{X} \mapsto \mathbf{AX} - \mathbf{XB}$	Definition 12.10.1
$\mathbf{B} - \lambda\mathbf{A}$	a matrix pencil; its generalized eigenvalues solve $\mathbf{Bx} = \lambda\mathbf{Ax}$	Definition 13.4.1

24.-1.9 Orders, convexity, and nonnegative matrices

Symbol	Meaning	Defined at
$\mathbf{x} \geq \mathbf{0}$, $\mathbf{A} \geq \mathbf{O}$, $\mathbf{A} > \mathbf{O}$	the **entrywise** order; a positive matrix has every entry positive	Definition 19.1.1
$\mathbb{R}^m_{\geq 0}$	the nonnegative orthant	Definition 18.5.1
$\mathbf{x} \prec \mathbf{y}$, $\mathbf{x} \prec_w \mathbf{y}$, $\mathbf{x}^{\downarrow}$	majorization, weak majorization, decreasing rearrangement	Definition 17.8.1
$\mathrm{conv}\, S$, $\mathrm{aff}\, S$	convex hull; affine hull, a coset of a subspace	Definition 18.1.4, Definition 18.1.5
$[x, y]$	the segment from x to y	Definition 18.1.1
Δ_n	standard simplex: n nonnegative weights adding to 1	Definition 18.1.2
$\mathrm{int}\,(S)$, $\overline{S}$	interior and closure in a finite-dimensional real space	Definition 18.3.2, Definition 18.1.6
$P_C(x)$, $d(x, C)$	nearest point of a closed convex set, and the distance to it	Theorem 18.3.1
p_K, h_K, K°	gauge, support function, and polar set of a convex set	Definition 18.4.1, Definition 18.4.2, Definition 18.4.3
K^*	dual cone, inside V itself	Definition 18.5.3
$\mathrm{cone}(a_1, \ldots, a_m)$	finitely generated cone	Definition 18.5.2
$\mathrm{epi}\, f$	epigraph, $\{(x, s) : s \geq f(x)\}$	Definition 18.10.1
$\mathrm{ext}\, C$	extreme points of a convex set	Definition 18.8.1, Definition 18.8.2

Symbol	Meaning	Defined at
$\mathcal{D}_n$, Ω_n	real density matrices; doubly stochastic matrices	Definition 18.8.3, Definition 18.8.4
$v(\mathbf{A})$	value of the matrix game with payoff matrix $\mathbf{A}$	Definition 18.7.2, Definition 18.7.1
$\mathbf{A}\#\mathbf{B}$	geometric mean of two positive definite matrices	Definition 21.11.1
$L_f(t_1, \ldots, t_n)$	Loewner matrix of divided differences	Definition 21.9.2, Definition 21.9.1
$G(\mathbf{A})$	directed graph of a square matrix: an edge $j \to i$ when $a_{ij} \neq 0$	Definition 19.2.1
irreducible, primitive	no proper invariant coordinate subspace; some power is positive	Definition 19.2.3, Definition 19.5.1
h, $\omega = e^{2\pi i/h}$	period of an irreducible nonnegative matrix, and the root of unity it names	Definition 19.5.2
$\underline{r}_{\mathbf{A}}(\mathbf{x})$, $\overline{r}_{\mathbf{A}}(\mathbf{x})$	lower and upper Collatz–Wielandt functions; the bars are not conjugation	Definition 19.4.1
Perron root and vectors	$\rho(\mathbf{A})$, the right vector $\mathbf{v} > \mathbf{0}$ with $\mathbf{1}^\top \mathbf{v} = 1$, the left vector $\mathbf{w}$ with $\mathbf{w}^\top \mathbf{v} = 1$	Definition 19.1.2, Definition 19.3.1
$\mathbf{Q}$, $\mathbf{R}$, $\mathbf{N}$	transient block, absorbing block, and fundamental matrix of an absorbing chain	Definition 19.6.1
$\mathbf{G}$, $\mathbf{H}$, α	Google matrix, raw link matrix, damping factor	Definition 19.8.2, Definition 19.8.1
$\mathbf{K}_n$	the second-difference matrix, a nonsingular M-matrix	Definition 19.9.2, Definition 19.9.1
stochastic matrix, steady state	column sums 1 and nonnegative entries; a fixed probability vector	Definition 9.11.1, Definition 9.11.2

24.-1.10 Algebras, groups, and representations

Symbol	Meaning	Defined at
A, 1_A	an associative algebra over F and its unit; plain italic, never bold	Definition 14.12.1
$\mathcal{T}_n$, $\mathcal{N}$	the upper triangular matrices in $M_n(F)$; its ideal of strictly upper triangular ones	Example (Six subalgebras)
$F\langle S\rangle$	the subalgebra generated by a set: the span of the words in S	Definition 23.1.4
$F[T]$, $F[\mathbf{A}]$	the algebra of polynomials in one operator or matrix, of dimension $\deg m_T$	Proposition 23.1.2
$F[G]$, δ_g	group algebra of a finite group and its basis of point masses	Definition 23.1.5
L_a	left multiplication by a; the regular representation is $a \mapsto L_a$	Theorem 23.1.3
S', S'', $Z(A)$	commutant, double commutant, center	Definition 23.1.6, Proposition 23.1.5
C_j, L_W, R_U	the column ideals of $M_n(F)$, and the left and right ideals of a subspace	Definition 23.2.1, Proposition 23.2.4

Symbol	Meaning	Defined at
$A/\mathcal{I}$, π	quotient algebra by a two-sided ideal, and the quotient homomorphism	Definition 23.2.2
$\operatorname{rad} A$	radical of an algebra: killed by every irreducible representation	Definition 23.3.5
av, an A-space	an action of A on V, that is a homomorphism $A \to \operatorname{End}(V)$	Definition 23.3.2
simple, semisimple	an A-space with no proper nonzero invariant subspace; a direct sum of such	Definition 23.3.3, Definition 23.3.4
$\operatorname{End}_A(V)$, $\operatorname{Hom}_A(V,W)$	equivariant maps, $f(av) = af(v)$; the subscript is never dropped	Definition 23.4.1
$A(\mathcal{F})$	the subalgebra generated by a family of operators	Lemma 23.4.2
$v\varphi$	the rank-one operator $x \mapsto \varphi(x)v$	Theorem 23.5.2
e_i	orthogonal idempotents of a product of algebras: italic, never bold	Definition 23.6.1
ρ, $\chi_\rho(g) = \operatorname{tr}\rho(g)$	a representation of a group and its character	Definition 23.7.1, Definition 23.8.2
irreducible, equivalent	no proper nonzero subrepresentation; conjugate by one isomorphism	Definition 23.7.4, Definition 23.7.3
$\operatorname{Cl}(G)$, $\langle f, f' \rangle$	class functions on G, and their averaged inner product	Definition 23.8.3, Definition 23.8.4
V^G	the fixed space of a representation	Lemma 23.8.4
$\langle g \rangle$, $G \times H$	the cyclic subgroup generated by g; a direct product of groups	Definition 23.9.1, Definition 23.9.2
$\widehat{G}$, $\widehat{f}$, ev_g	dual group of a finite abelian group, Fourier transform, evaluation at g	Definition 23.9.3, Definition 23.9.5
$\mathbf{U}$	character matrix, $(\mathbf{U})_{mj} = \chi_j(g_m)/\sqrt{n}$, indices from 0	Definition 23.9.4
$\mathbf{C}$, $\mathbf{F}$	circulant matrix; the Fourier matrix	Definition 12.9.1, Theorem 12.9.3
$\mathbf{c} * \mathbf{d}$	cyclic convolution, subscripts read modulo n	Definition 12.9.1

24.-1.11 Geometry

Symbol	Meaning	Defined at
$\mathcal{A}, \mathcal{B}, \mathcal{C}$	affine spaces, plain calligraphic	Definition 22.1.1
P, Q, R, $\overrightarrow{PQ}$, $P + v$	points of an affine space, **not bold**; the vector between two points; translation of a point	Definition 22.1.1
$\mathbb{A}^n(F)$	affine n-space: F^n with its origin forgotten	Definition 22.1.1
θ_O	the choice-of-origin bijection $\mathcal{A} \to V$, $P \mapsto \overrightarrow{OP}$	Proposition 22.1.3
$\vec{f}$, $\widehat{f}$	linear part of an affine map; its block matrix, homogenizing coordinate **last**	Theorem 22.1.5, Proposition 22.1.7
$\operatorname{Aff}(\mathcal{A})$, τ_v	the affine group, and translation by v	Theorem 22.1.6

Symbol	Meaning	Defined at
$\vec{A}$, $A \vee B$, $A \parallel B$	direction space of a flat; the join; parallelism	Definition 22.2.1, Definition 22.2.2, Definition 22.2.3
P_J	the face of a polyhedron cut out by making the rows $i \in J$ tight	Proposition 22.3.1, Definition 22.3.1
$\mathrm{E}(n)$, $t_{\mathbf{b}}$, $\mathrm{Fix}(f)$	the Euclidean group, translation by $\mathbf{b}$, fixed set of a motion	Definition 22.4.1
$\mathbf{R}_\theta$, $\mathbf{M}_\theta$	plane rotation and plane reflection matrices	Theorem 11.7.6
$\widetilde{\mathbf{A}}$, $\tilde{\mathbf{x}}$	big matrix and big vector of a quadric, with $Q(\mathbf{x}) = \tilde{\mathbf{x}}^\top \widetilde{\mathbf{A}} \tilde{\mathbf{x}}$	Definition 22.5.1
$\mathbb{P}(V)$, $\mathbb{P}^n(F)$	projective space of V, of dimension $\dim V - 1$; $\mathbb{P}(F^{n+1})$	Definition 22.6.1
$[\mathbf{x}] = [x_0 : \cdots : x_n]$	homogeneous coordinates; coordinates run from 0	Definition 22.6.2
U_0, U_j, H_∞, ι	the standard affine chart, the other charts, the hyperplane at infinity, the chart bijection	Theorem 22.6.2
$S \vee T$, $S \cap T$	join and meet of projective subspaces	Proposition 22.6.5, Definition 22.6.3
$[T]$, $[\mathbf{A}]$	the projectivity induced by an isomorphism or an invertible matrix	Definition 22.7.1
$\mathrm{PGL}(V)$, $F^\times$	projective linear group; the multiplicative group of the field	Theorem 22.7.3
$(A, B; C, D)$	cross-ratio of four points: itself a point of $\mathbb{P}^1(F)$, not a scalar	Definition 22.7.3
S^0	projective annihilator, Chapter 5's U^0 read in $\mathbb{P}(V^*)$	Theorem 22.8.3, Definition 22.8.1
o	a chosen representative vector of a projective point	Lemma 22.9.1
$Z(q)$, P^β	zero set of a quadratic form; the polar of a point with respect to a conic	Definition 22.10.1, Definition 22.10.2
p_{ij}, $[L]$, $\mathcal{K}$	Plücker coordinates of a line, the point they form, and the Klein quadric	Definition 22.10.3
vol	volume: of a parallelepiped it is $\lvert\det\rvert$; extended to k-dimensional volume	Definition 7.9.1, Definition 13.3.1
$(\mathrm{In}\,\mathbf{A}, \mathrm{In}\,\widetilde{\mathbf{A}})$	the pair of inertia triples of a real quadric, an invariant of its affine equivalence class	Definition 22.5.2, Definition 17.10.1

24.-1.12 Computation

Symbol	Meaning	Defined at
u, fl, $\mathcal{R}$, δ	unit roundoff, the rounding map, the representable numbers, one rounding	Definition 24.1.1
$\gamma_k = ku/(1-ku)$	the accumulated-rounding constant	Lemma 24.1.1
$\widehat{\mathbf{x}}$, $\Delta\mathbf{A}$	a computed object; a data perturbation produced by an algorithm	Definition 24.1.2, Definition 24.1.3

Symbol	Meaning	Defined at
flop, g_n	one arithmetic operation; the growth factor of elimination	Definition 24.2.1, Definition 24.13.1
$\mathcal{K}_k(\mathbf{A}, \mathbf{b})$, $\theta_1, \ldots, \theta_k$	Krylov space $\operatorname{span}(\mathbf{b}, \mathbf{Ab}, \ldots, \mathbf{A}^{k-1}\mathbf{b})$; the Ritz values from it	Definition 24.6.1, Definition 24.6.2
Ł, $Ł_0$, $\mu_2(Ł)$	graph Laplacian $\mathbf{D} - \mathbf{A}$, the reduced Laplacian, the Fiedler value	Definition 24.11.1, Definition 24.11.2
$\mathbf{X}$, $\mathbf{X}_c$, $\mathbf{S}$	data matrix, its centered form, the sample covariance	Definition 24.10.1, Definition 24.10.2
$\mathbf{M}$, $\mathbf{K}$, ω_j	mass matrix, stiffness matrix, natural frequencies	Definition 24.9.2
$\mathbf{W}_n$, $\mathbf{P}_n$, $\mathbf{D}_m$, $\mathbf{B}_n$	unnormalized transform matrix, even-odd permutation, twiddle matrix, butterfly	Definition 24.12.1, Definition 24.12.2
ω	the exponent of matrix multiplication	Definition 24.13.2

24.-1.13 Where the book overrides itself

Twenty-four chapters do not have twenty-four alphabets. Several letters carry more than one meaning, and a few English words carry more than one definition. In every case the local meaning is declared where it is used, and in every case there is a tell — an argument, a subscript, a typeface — that decides which one is meant. The rows below list the clashes worth knowing before you meet them.

Symbol or word	The meanings	How to tell
U	a generic upper triangular matrix (Definition 0.9.5, Definition 3.6.1); a unitary matrix, in the Schur and singular value factorizations (Corollary 12.1.3, Definition 13.9.1); the character matrix of a finite abelian group (Definition 23.9.4)	the chapter: triangular in Chapters 3 and 23, unitary from Chapter 12 on, the character matrix only in Chapter 23 §09
V	the right factor of a singular value decomposition, $\mathbf{A} = \mathbf{U\Sigma V}^*$ (Theorem 13.9.1); the Vandermonde matrix $\mathbf{V}(x_1, \ldots, x_n)$ (Theorem 7.8.1); one of the two matrices invertible over $F[x]$ in $\mathbf{B} = \mathbf{UAV}$ (Definition 10.7.2). Plain V is a vector space	the Vandermonde matrix always carries its nodes; the typeface for the space, and Chapter 10 §07 is the one place **V** is invertible only over $F[x]$
irreducible	a polynomial with no nontrivial factorization (Definition 6.3.1); a nonnegative matrix with no invariant coordinate subspace (Definition 19.2.3); an A-space or a representation with no proper nonzero invariant subspace (Definition 23.3.3, Definition 23.7.4)	what the word is applied to. The three notions are unrelated
e_i versus $\mathbf{e}_i$	italic e_i is an idempotent of a product of algebras (Definition 23.6.1); bold $\mathbf{e}_i$ is the i-th standard basis vector (Example (Standard Bases))	the typeface, and that is the whole reason Chapter 23 chose italic

Symbol or word	The meanings	How to tell
G	Gram matrix (Definition 11.2.3); Givens rotation $\mathbf{G}(i, j; c, s)$ (Definition 11.8.2); Google matrix (Definition 19.8.2); the alternating Gram matrix of the Plücker form (Definition 22.10.3). Plain $G(\mathbf{A})$ is the directed graph of a matrix (Definition 19.2.1)	the Givens rotation always carries its four arguments; the graph is plain italic with its matrix in brackets
H, **K**	Hessian $\mathbf{H}_f(\mathbf{a})$ (Definition 14.6.2); Householder reflection $\mathbf{H}_\mathbf{w}$ (Definition 11.8.1); the Hermitian and skew parts of a matrix (Lemma 20.2.3); the raw link matrix (Definition 19.8.1); the second-difference matrix $\mathbf{K}_n$ (Definition 19.9.2); the stiffness matrix (Definition 24.9.2). Calligraphic $\mathcal{K}_k(\mathbf{A}, \mathbf{b})$ is a Krylov space (Definition 24.6.1) and $\mathcal{K}$ the Klein quadric (Definition 22.10.3); blackboard $\mathbb{H}$ is the quaternions (Definition 14.12.3)	the Hessian and the Householder reflection always carry their argument, the second-difference matrix always its size $\mathbf{K}_n$, the Krylov space its subscript and arguments
Ł	the lower triangular factor of an elimination (Definition 3.6.1); the graph Laplacian (Definition 24.11.1); the Lax matrix (Definition 24.8.1). Plain L_a is left multiplication in an algebra (Theorem 23.1.3) and L_f a Loewner matrix (Definition 21.9.2); calligraphic $\mathcal{L}(V, W)$ is a space of maps (Definition 2.4.1)	Chapter 24 §§08 and 11 each declare their use at the head of the section; the plain letters carry a subscript
R, r	the **R** of a QR factorization; the plane rotation $\mathbf{R}_\theta$ (Theorem 11.7.6); the rotation through $2\pi/n$ in Chapter 23; the absorbing block of a Markov chain (Definition 19.6.1). Plain $R_\mathbf{A}(\mathbf{x})$ is the Rayleigh quotient (Definition 17.1.1) and $r_i(\mathbf{A})$ a Gershgorin radius (Definition 20.1.1)	the Rayleigh quotient is plain italic and takes a vector; the Gershgorin radius is plain and takes an index
Q	the orthogonal factor of a QR factorization; the orthogonal linear part of a Euclidean motion (Definition 22.4.1); the quarter turn in Chapter 23; the transient block of an absorbing chain (Definition 19.6.1). Blackboard $\mathbb{Q}$ is the rationals (Definition 0.3.1)	the chapter, and the blackboard face for the field
S, $\mathcal{S}$	a congruence or a similarity matrix; a Schur complement in Chapter 24 §02; the sample covariance (Definition 24.10.2); the cyclic shift in Chapter 12 §09. Calligraphic $\mathcal{S}_{\mathbf{A},\mathbf{B}}$ is the Sylvester operator (Definition 12.10.1)	each section of Chapter 24 says which it means; the Sylvester operator carries both its matrices
N, $\mathcal{N}$	the fundamental matrix $(\mathbf{I} - \mathbf{Q})^{-1}$ of an absorbing chain (Definition 19.6.1); an incidence matrix, with Ł $= \mathbf{N}\mathbf{N}^\top$ (Definition 3.7.1); the strictly upper triangular matrices $\mathcal{N}$ (Example (Six subalgebras)). Blackboard $\mathbb{N}$ is the natural numbers	the calligraphic face for the algebra, the blackboard face for the number set
J	the all-ones matrix (Example (Orthogonally diagonalizing a 3 by 3 matrix)); the Jordan block $\mathbf{J}_k(\lambda)$ (Definition 10.3.1)	the Jordan block always carries its size and its eigenvalue

Symbol or word	The meanings	How to tell
$\mathbf{C}$	the companion matrix $\mathbf{C}(p)$ (Definition 10.5.3); the real Jordan block $\mathbf{C}_k(\lambda)$ (Definition 10.4.1); a circulant (Definition 12.9.1). Plain C and C_b^a are contractions (Definition 15.4.1) and C_{ij} a cofactor (Definition 7.6.1); blackboard $\mathbb{C}$ is the complex numbers	the companion matrix always carries its polynomial, the Jordan blocks their size and eigenvalue; a bare $\mathbf{C}$ in Chapter 23 is a circulant
$\mathbf{W}$, W	the Wielandt matrix $\mathbf{W}_n$ (Theorem 19.5.6); the unnormalized transform matrix $\mathbf{W}_n$ of Chapter 24 §12; the accumulated orthogonal factor $\mathbf{W}_k$ of Chapter 24 §08. Plain W is a subspace, $W(\mathbf{A})$ the numerical range (Definition 20.2.1), and W_n the zero-sum matrices of Chapter 19 §07 (Lemma 19.7.1)	the numerical range always carries its matrix as an argument; the subspace never carries one
Θ, θ	the isomorphism $\mathbb{C} \to \mathbb{R}^2$ (Example (The Standard Isomorphisms)); the basis-dependent $V \to V^*$ (Theorem 5.3.2); the natural isomorphism $V^* \otimes W \to \mathcal{L}(V, W)$ (Theorem 15.3.9); the diagonal matrix of principal angles (Definition 20.8.1). Lowercase: the angle between vectors (Definition 11.1.5), the choice-of-origin map θ_O (Proposition 22.1.3), the Ritz values $\theta_1, \ldots, \theta_k$ (Definition 24.6.2)	the principal-angle Θ appears only inside $\sin\Theta$ and $\cos\Theta$; lowercase θ is an angle unless the section says otherwise, and the choice-of-origin map carries the point O
Π, π	bold $\mathbf{\Pi}$ is the limit of a convergent stochastic matrix (Theorem 10.10.4); plain Π is a plane in Chapter 22 §§04, 08, 09 and the quotient map $\mathrm{GL}(V) \to \mathrm{PGL}(V)$ in §07; π is the quotient map $X \to X/\!\sim$ of a set (Definition 0.5.5), a quotient homomorphism of algebras (Definition 23.2.2), and, locally, a slot projection	the typeface, and that no section of Chapter 22 uses both meanings
Φ, Ψ	a symmetric gauge function and its dual (Definition 21.3.1, Definition 21.3.3); the two isomorphisms of Chapter 23 §09 (Definition 23.9.5)	the argument: a gauge eats a real vector, Chapter 23's maps eat an element of a group algebra
Ω, ω	plain Ω_n is the doubly stochastic matrices (Definition 18.8.4), bold $\mathbf{\Omega}_{2m}$ the standard alternating matrix (Definition 14.10.2). Lowercase ω is a root of unity (Definition 19.5.2, Definition 24.12.1), an angular frequency (Definition 24.9.2), the exponent of matrix multiplication (Definition 24.13.2), and a differential form being wedged (Definition 15.10.1)	the typeface for Ω; for ω, the section, each of which declares its use
ρ	spectral radius (Definition 16.4.1); a representation of a group or an algebra (Definition 23.7.1)	the argument: a matrix for the radius, a group element for the representation

Symbol or word	The meanings	How to tell
σ	a permutation (Definition 7.3.2); a singular value $\sigma_i(\mathbf{A})$ (Definition 13.9.1)	the singular value always carries an index and a matrix
λ, μ	a scalar or an eigenvalue throughout; also the linear-part homomorphism $\mathrm{E}(n) \to \mathrm{O}(n)$ (Definition 22.4.1) and the Fiedler value μ_2 (Definition 24.11.2)	Chapter 22 §04 is the one place λ is not a scalar, and it says so
$\lvert\cdot\rvert$	modulus of a complex number (Definition 0.6.2); cardinality of a set; the matrix absolute value $(\mathbf{A}^*\mathbf{A})^{1/2}$ (Definition 13.2.1); the **entrywise** absolute value in Chapter 19 (Definition 19.1.1)	what sits inside. Chapter 19 states its override, and the Chapter 13 meaning does not appear there
Λ	plain $\Lambda^k V$ is an exterior power (Definition 15.8.2); bold $\mathbf{\Lambda}(\lambda)$ is a rotation-scaling block (Definition 10.4.1)	the typeface, and the block always carries its eigenvalue
δ	Kronecker delta δ_{ij} (Definition 0.9.3); a point mass δ_g in a group algebra (Definition 23.1.5); a basis element δ_s of a free vector space (Definition 15.2.3); one rounding error (Definition 24.1.1)	the Kronecker delta always carries two **numerical** subscripts
$\perp$	orthogonal complement $U^\perp$ (Definition 11.3.1); the complement $U^{\perp_\beta}$ for a bilinear form (Definition 14.1.3); orthogonal direct sum $U \perp W$ of a form (Definition 14.7.3); a polarity $S^\perp$ in projective space (Theorem 22.8.3)	the unadorned $U^\perp$ is always the inner-product one
$\cong$, $\sim$, $\simeq$, $\approx$	isomorphic spaces (Definition 2.5.3) or, between spaces carrying forms, isometric (Definition 14.8.1); similar matrices (Definition 4.2.2); congruent matrices (Definition 14.2.1); equivalent over $F[x]$ (Definition 10.7.2), or "approximately equal" in prose	what stands on either side; the four relation symbols are never interchanged
brackets	$[v]_{\mathcal{B}}$ a coordinate vector (Definition 1.5.2); $[T]^{\mathcal{C}}_{\mathcal{B}}$ a matrix of a map (Definition 4.1.1); $[\mathbf{A}, \mathbf{B}]$ a commutator (Definition 8.5.1); $[x, y]$ a segment (Definition 18.1.1); $[\mathbf{x}]$ a projective point (Definition 22.6.2) and $[T]$ a projectivity (Definition 22.7.1); $F[x]$ and $F[G]$ rings and algebras (Definition 0.8.2, Definition 23.1.5)	the number of arguments and whether they are bold: a segment takes two bold vectors, a commutator two bold matrices

Symbol or word	The meanings	How to tell
the superscript of $[\,\cdot\,]_{\mathcal{B}}^{\mathcal{C}}$	for a linear map it is the **output** basis, $[T]_{\mathcal{B}}^{\mathcal{C}}$ (Definition 4.1.1); for a bilinear form it is the basis read into the **second slot**, $[\beta]_{\mathcal{B}}^{\mathcal{B}}$ (Definition 14.1.2)	what sits inside the brackets. The two readings are the same rule — the basis on the second slot — since a map's second slot is its output; a form on $V \times V$ always gives both slots the same basis, so its two labels agree, and neither is ever left off
T	a linear map throughout; the transpose $\mathbf{A}^{\top}$; Chebyshev polynomials T_n (Definition 24.7.4); the tensor algebra $\mathrm{T}(V)$ (Definition 15.6.2) and mixed tensors $T_q^p(V)$ (Definition 15.4.2); a T-transform (Definition 21.1.1)	the tensor algebra is upright and takes a space; the Chebyshev polynomial and the T-transform carry subscripts
P, p	orthogonal projection P_U (Definition 11.3.2); Legendre polynomials P_n (Definition 11.10.1); a point of an affine space (Definition 22.1.1); the nearest-point map P_C (Theorem 18.3.1); a face P_J (Proposition 22.3.1); the even-odd permutation $\mathbf{P}_n$ (Definition 24.12.2); characteristic polynomial p_T (Definition 7.10.1); a gauge p_K (Definition 18.4.1)	a subspace subscript means a projection, an integer subscript a polynomial; bold means a matrix
Z	the cyclic subspace $Z(v;T)$ (Definition 10.5.1); the center $Z(A)$ of an algebra (Definition 23.1.6); the zero set $Z(q)$ of a quadratic form (Definition 22.10.1); a Z-matrix (Definition 19.9.1)	the argument decides: a vector and an operator, an algebra, or a form
$\mathbf{E}$, E	the perturbation in $\mathbf{A}+\mathbf{E}$; a matrix unit $\mathbf{E}_{ij}$ (Example (Standard Bases)); an elementary matrix (Definition 3.4.1); an eigenspace $E_\lambda(T)$ (Definition 9.2.2); the Euclidean group $\mathrm{E}(n)$ (Definition 22.4.1)	subscripts again: two indices give a matrix unit, an eigenvalue gives an eigenspace, a bare $\mathbf{E}$ is the perturbation
$\mathbf{D}$, D	a diagonal matrix (Definition 0.9.4); the elementary scaling $\mathbf{D}_i(c)$ (Definition 3.4.1); the twiddle matrix $\mathbf{D}_m$ (Definition 24.12.2); the degree matrix in $Ł = \mathbf{D} - \mathbf{A}$ (Definition 24.11.1); a Gershgorin disc $D_i(\mathbf{A})$ (Definition 20.1.1); density matrices $\mathcal{D}_n$ (Definition 18.8.3)	the Gershgorin disc is plain italic, since the bold letter is a diagonal matrix
$\mathbf{M}$, M	a block matrix, as in the Schur complement $\mathbf{M}/\mathbf{A}$ (Definition 8.3.1); the mass matrix (Definition 24.9.2); the plane reflection $\mathbf{M}_\theta$ (Theorem 11.7.6); a minor M_{ij} (Definition 7.6.1); the space $M_n(F)$ (Definition 0.9.1); an M-matrix (Definition 19.9.2)	plain italic with two indices is a minor; $M_n(F)$ always carries its size and field

Symbol or word	The meanings	How to tell
u	the unit roundoff (Definition 24.1.1); bold $\mathbf{u}$ is a vector	the typeface; Chapter 24 §01 warns about the pair
h	the period of an irreducible nonnegative matrix (Definition 19.5.2); a support function h_K (Definition 18.4.2); a mesh width in Chapter 19 §09	the support function always carries its convex set
simple	a simple algebra (Definition 23.3.1); a simple A-space (Definition 23.3.3); a simple eigenvalue, of multiplicity one (Definition 20.11.1); a simple tensor (Definition 15.2.2)	what the word is applied to
equivalent	equivalent matrices $\mathbf{B} = \mathbf{QAP}$ (Definition 4.3.1); equivalence over $F[x]$ (Definition 10.7.2); equivalent norms (Definition 16.2.1); equivalent representations (Definition 23.7.3); affine equivalence of quadrics (Definition 22.5.2)	the objects being compared
positive	$\mathbf{A} > \mathbf{O}$ entrywise, every entry positive (Definition 19.1.1); $\mathbf{A} \succ \mathbf{O}$ positive definite (Definition 13.1.1)	the symbol: $>$ is entrywise, $\succ$ is the Loewner order. A positive matrix need not be positive definite, nor conversely
radical, **contraction**	radical of a bilinear form (Definition 14.1.3) or of an algebra (Definition 23.3.5); contraction of tensor slots (Definition 15.4.1) or a matrix of norm at most one (Definition 21.8.1)	the argument, in both cases
signature	the pair (n_+, n_-), and also the single integer $n_+ - n_-$	Definition 14.5.1 says which is meant where the word appears